Sujal Banshilal Meshram

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.22
Dasmesh Public School, Faridkot	Class XII (CBSE)	2023	95.00%
Montessori Cambridge School, Pathankot	Class X (CBSE)	2021	94.17%

WORK EXPERIENCE

Google SWE Intern - AI Developer Tools

- Developed a full-stack AI diagnostic console using Java, Kotlin, Protocol Buffers, and AngularDart to trace agentic AI workflows, inspect session logs, rerun agents with prompt overrides, and surface inline source citations.
- Built a Python LLM evaluation pipeline integrating Google Sheets, Agent Orchestration Service, F1 DB, and an AI evaluation framework to benchmark agent responses against human-verified golden datasets.

PROJECTS

Central Communication and Intelligent Reminder Platform

- Built a multi-channel communication platform using FastAPI, async Python, MongoDB, Next.js, and TypeScript with priority-based Email, SMS, and WhatsApp dispatch.
- Integrated Gemini tool-calling, Sentence-BERT segmentation, Celery/Redis, Kafka, and atomic MongoDB job claiming.

JADE: Distributed/HPC DSL

- Designed JADE, a domain-specific language for distributed computing, with role/task abstractions, communication primitives, static type checking, and deadlock checks.
- Built the lexer-to-C++/MPI compiler pipeline with automatic role-to-rank mapping and demonstrated distributed matrix multiplication/Sudoku workloads.

Online Auction System

- Built a full-stack auction platform using Go, React, and PostgreSQL with JWT authentication, RBAC, automated bidding, and real-time updates.
- Handled concurrent bids using transactional locking and automated deadlines, email notifications, and bid processing.

Real-Time Chat App

- Developed a real-time chat application using the **MERN stack**, with a Node.js/Express backend, MongoDB Atlas, JWT authentication, and React frontend.
- Integrated **Socket.IO** for real-time messaging and online presence indicators, with **Zustand** for lightweight client-side state management.

RISC-V Simulator

- Developed a RISC-V RV64I assembler and cache simulator in C++ with an interactive CLI supporting execution, single-stepping, breakpoints, and register/memory inspection.
- Implemented configurable direct-mapped, set-associative, and fully associative caches with FIFO, LRU, and Random replacement and write-back/write-through policies.

Fraud Detection using GANs

- Built GAN-based semi-supervised fraud detection in PyTorch, training on normal transactions with custom BiGAN and AnoGAN architectures.
- Compared single-pass BiGAN and latent-optimization AnoGAN using reconstruction/feature-matching anomaly

scores; achieved 0.938 and 0.958 ROC-AUC respectively.

Infinite Precision Arithmetic Library

- Designed C++ Integer and Float classes using normalized digit-string representations with explicit sign handling, robust parsing, and configurable floating-point precision.
- Overloaded arithmetic, comparison and stream operators for an STL-style interface.

RELEVANT COURSES

Theory: Algorithms, Data Structures, Theory of Computation, Approximation Algorithms, Distributed Computing

Mathematics: Probability and Random Variables, Linear Algebra, Calculus, Statistics, Transform Techniques, Linear Optimization, Convex Optimization

AI & Data: Deep Learning, Natural Language Processing, Machine Learning, Artificial Intelligence, Data Mining, Fraud Analytics, Multi-Agent Systems

Systems: Computer Architecture, Operating Systems, Computer Networks, Parallel and Concurrent Programming, Compilers

SKILLS

Languages: C/C++, Python, Go, Dart, Java, JavaScript

Web: Node, Express, React, Socket.IO, HTML, CSS

DBMS: PostgreSQL, MySQL, MongoDB

SCHOLASTIC ACHIEVEMENTS

Actively participated in **Codeforces** contests, reaching a **Specialist** rating (1548)

Qualified the prestigious two-stage **National Talent Search Examination** (NTSE) securing a national-level scholarship

EXTRACURRICULAR

NSS Volunteer: Participated in NSS-led initiatives for **Campus Improvement & Community Service**